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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 05/21/2004 |
| **PI Name:** | Shapiro, Irwin Ph.D. | | |
| **Project Title:** | Test of the Equivalence Principle in an Einstein Elevator | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Gravitational & relativistic physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Cambridge | **State:** | MA |
| **Zip Code:** | 02138 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 04/30/2001 | **End Date:** | 04/30/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Glashow, Sheldon ( Boston University Boston MA )<br>Iafolla, Valerio ( Institute of Interplanetary Space Physics (IFSI) Rome Italy )<br>Lorenzini, Enrico ( Smithsonian Astrophysical Observatory Cambridge MA )<br>Cheimets, Peter ( Smithsonian Astrophysical Observatory Cambridge MA )<br>Cosmo, Mario ( Smithsonian Astrophysical Observatory Cambridge MA )<br>Finkelstein, Nancy ( Smithsonian Astrophysical Observatory Cambridge MA )<br>Schneps, Matthew ( Smithsonian Astrophysical Observatory Cambridge MA )<br>Nozzoli, Sergio ( Institute of Interplanetary Space Physics (IFSI) Rome Italy )<br>Santoli, Francesco ( Institute of Interplanetary Space Physics (IFSI) Rome Italy ) | | |
| **Grant/Contract No.:** | None | | |
| **Performance Goal No.:** | 1.24 | | |
| **Performance Goal Text:** | To explore the frontiers of physics anbd unlock the mysteries of the universe obscured by gravity | | |

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| **Task Description:** | The goal of this grant is to carry out the flight definition of a system for testing repeatedly the Equivalence Principle with an accuracy significantly improved with respect to the state of the art. The experimental technique utilizes a differential accelerometer (detector) in free fall inside an evacuated capsule released from a stratospheric balloon. The detector free falls, while rotating slowly, inside the co-moving shielding capsule which drops up to 4 km while providing a very high level of isolation from external noise sources. The detector measures very accurately the separation (and hence the differential rate of fall) of a pair of test masses of different composition. The experiment can be repeated periodically and, consequently, allow for fine tuning of the experiment, improvements of the detector, and testing of different materials. The ultimate objective of the experiment is to test the principle that lies at the foundation of the currently-accepted gravitational theory (general relativity), that is, the equivalence between the mass seen as a response to gravity and the mass seen as a repository of inertia. The goal of the flight experiments is to push the accuracy in testing this equivalence up to several parts in 10^15 or, similarly, in measuring the relative rate of fall of a pair of test masses of different composition (the sensing elements of the differential acceleometer) up to that accuracy. The target accuracy is about two orders of magnitude better than the present state of the art.<br>The flight definition activities covered by this grant will be as follows: (a) assessment of science requirements and the definition of engineering requirements for the experiment, (b) analysis of noise sources and their mitigation, (c) preliminary design of the free-fall system; (d) construction and laboratory testing of a prototype release mechanism; (e) analysis of the theoretical implications of testing the Equivalence Principle with an improved accuracy; and (f) an E/PO activity with the goal of conveying basic ideas of gravitational physics to young students and educators. In parallel to the tasks above, the group at the Institute of Space Physics (IFSI) in Rome, funded by the Italian Space Agency, will develop a prototype differential accelerometer and conduct key tests on the prototype accelerometer in the laboratory.<br>The Equivalence Principle is at the foundation of gravitational theories and thus affects our understanding of the fundamental forces in nature. A negative result of the test (non violation) will impose very stringent limits on gravitational theories while a positive result (violation) will likely imply the need for a fundamental revision in the theory of general relativity and, indeed, in other basic laws developed to describe nature. The results of the tasks to be carried out under this grant will define the scientific and technical requirements of the experiment, assess its scientific/technical soundness, and lay the groundwork for the balloon experiments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | The research activity during this fiscal year has focused on the following tasks: (a) an analysis of suitable proof masses and their gravity coupling to outer masses; (b) the development of a model to study the motion of the instrument package during free fall; (c) the study of the dynamics of the balloon and gondola before capsule release; (d) a preliminary design of the leveling mechanism that connects the capsule to the gondola; (e) the conceptual design of the liquid Helium cryostat that houses the experiment; and (f) the design of a new detector prototype by our non-US partners. In brief, we evaluated the shapes of sensing masses that have sufficiently low gravitational coupling with the capsule and that are suitable for a detector construction. The simulation model of the instrument package and detector dynamics has been instrumental in computing inertia characteristics of the instrument package that enable the separation of the dynamics-related noise from the expected signal. The study of the balloon and gondola dynamics has led to the derivation of requirements for the leveling mechanism that provides the capsule verticality before release. The model was also used to explore possible control strategies for attenuating the capsule oscillations before release. Moreover, our engineering department completed a preliminary design of the leveling mechanism and identified its principal components. Our subcontractor, Topsfield Engineering, delivered a conceptual design of the liquid Helium cryostat and evaluated the cost to design and build the cryostat. Based on the cryostat new concept, thermal analyses were conducted in house to assess the thermal gradients across the instrument package. Finally, our non-US partners have designed a new differential accelerometer prototype that has several features in common with the experiment detector. The new detector prototype will be used to conduct laboratory tests aimed at addressing critical issues related to the development of the sensor for the balloon experiments. |
| **Bibliography Type:** | Description: (Last Updated: 07/09/2007) |
| **Abstracts for Journals and Proceedings** | Iafolla, V.;Lucchesi, D.;Nozzoli, S.;Ravenna, M.;Santoli, F.;Shapiro, I.I.;Lorenzini, E.C.;Cosmo, M.L.; Ashenberg, J.;Cheimets, P.N.;Glashow, S. "General Relativity Accuracy Test (TEPEE/GReAT): New Configuration for the Differential Accelerometer" 35th COSPAR Scientific Assembly, Paris, France, July 2004<br>COSPAR Conference Abstracts , Jul-2004 |
| **Articles in Peer-reviewed Journals** | Ashenberg, J.;Lorenzini, E.C. "Analytical formulation of a complex mutual gravitational field" Classical and Quantum Gravity, Vol. 21, pp. 2089-2100. , Mar-2004 |